



LONDON AIRBNB

Market Analysis

A Data Analyst Portfolio Project

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Objective

This project analyzes London's Airbnb listings to understand pricing behavior, market structure, and host activity across the city.

- Which boroughs command the highest prices, and which have the most supply?
- How does room type (entire home vs private room) influence pricing?
- Does pricing relate to how often a listing is booked or reviewed?
- Is the market dominated by individual hosts or large-scale operators?

Data Source & Methodology

Source

Inside Airbnb
(insideairbnb.com)

Dataset

London listings.csv
92,799 raw records

Tools

Python, Pandas,
Matplotlib, Seaborn

Environment

Google Colab
(Jupyter notebook)

Methodology: raw listings were loaded, cleaned of missing/invalid values, capped for extreme price outliers, then explored through univariate and bivariate visual analysis.

Data Cleaning Summary

Raw Dataset

92,799 rows
19 columns

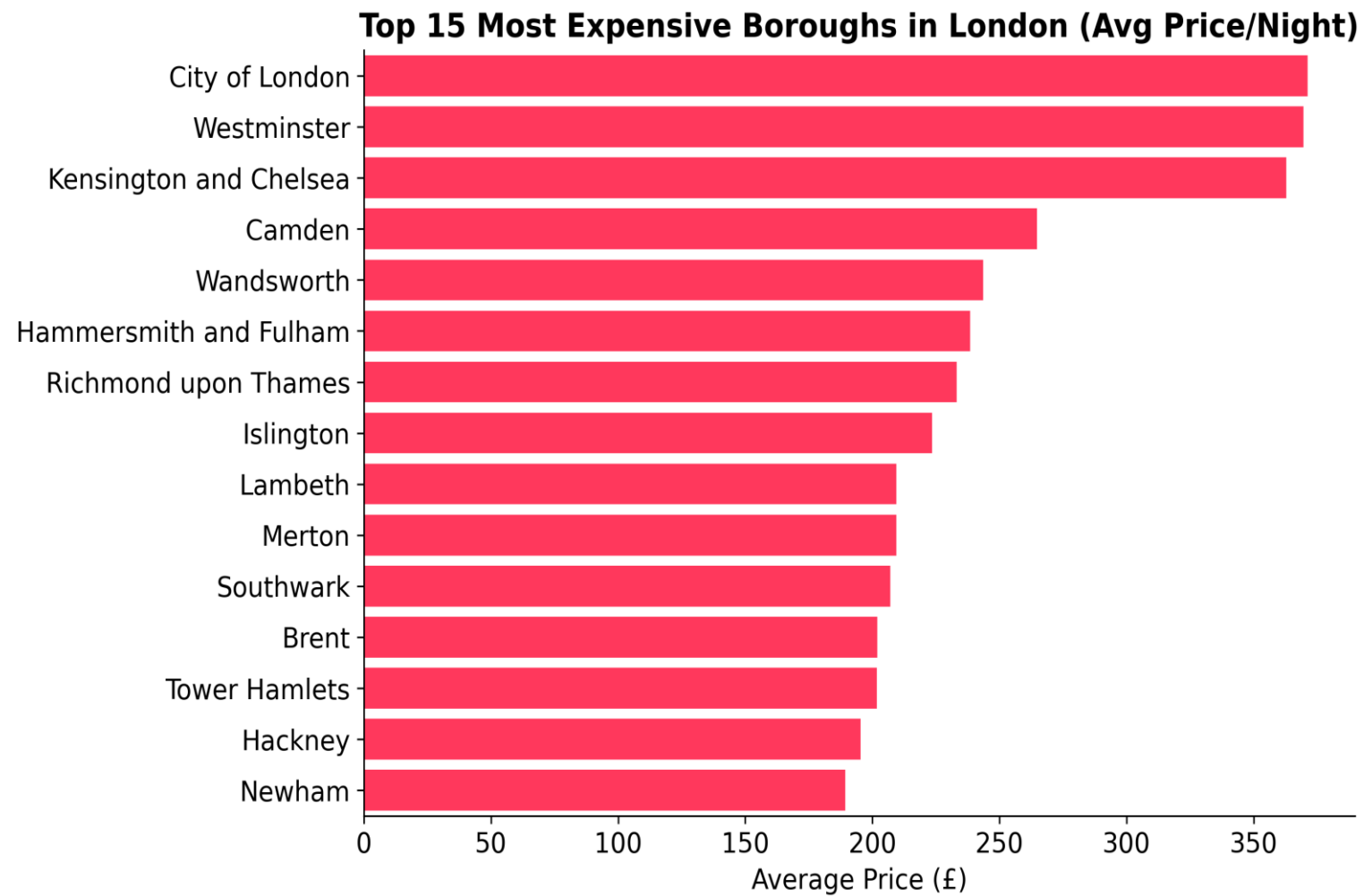
- Removed neighbourhood_group (100% empty)
- Removed license (100% empty)
- Dropped rows with missing price (~30,400)
- Capped price at the 99th percentile
- Converted last_review to date format

Cleaned Dataset

61,774 rows
17 columns

Every remaining column is fully populated or intentionally left blank where a listing genuinely has no reviews. This is the dataset used for all analysis in this deck.

Most Expensive Boroughs



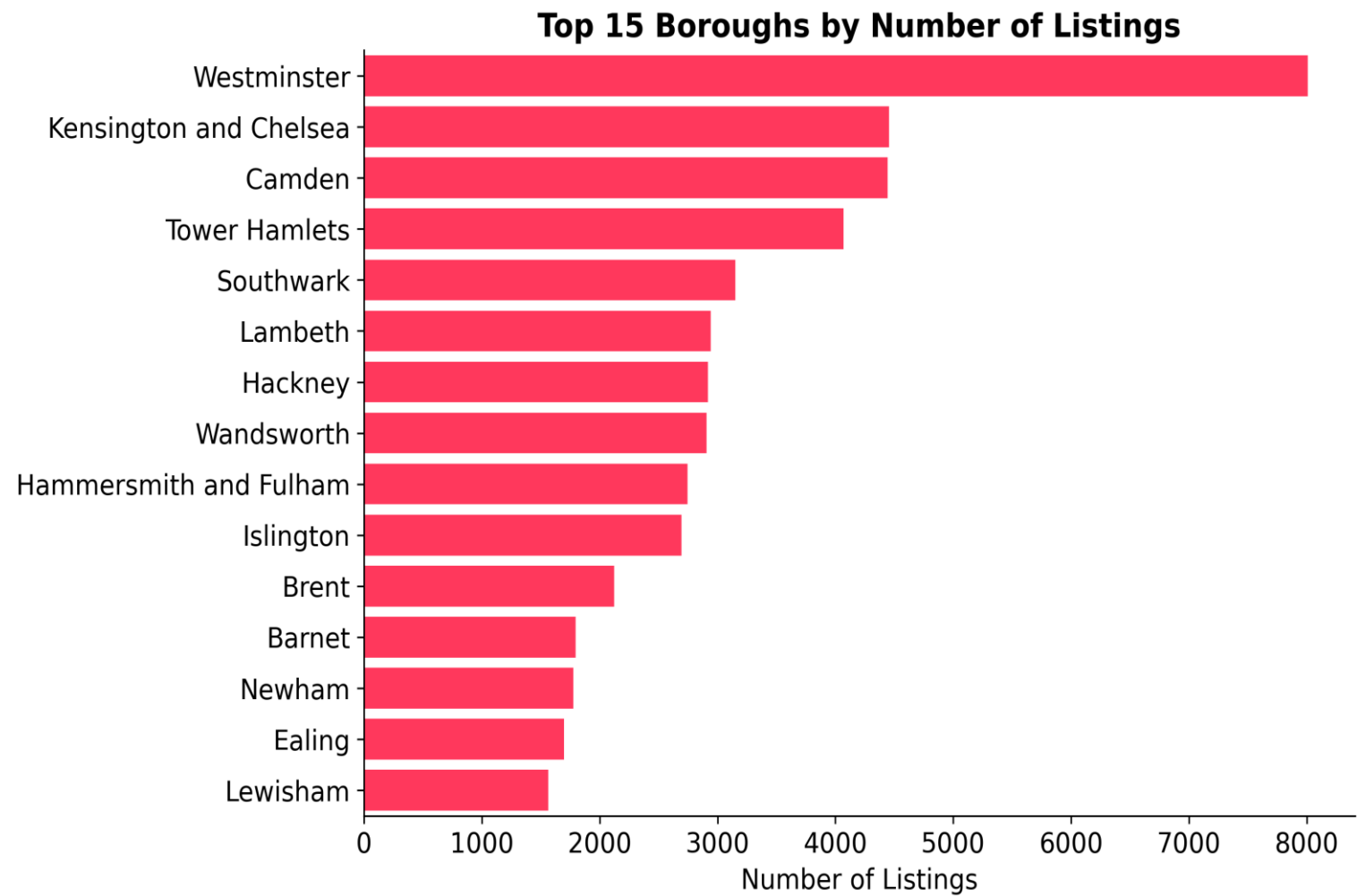
£371

Highest avg. price

Insight

City of London tops pricing at roughly £371/night on average, more than double the citywide median — driven by its financial-district, corporate-stay demand.

Supply by Borough



Westminster

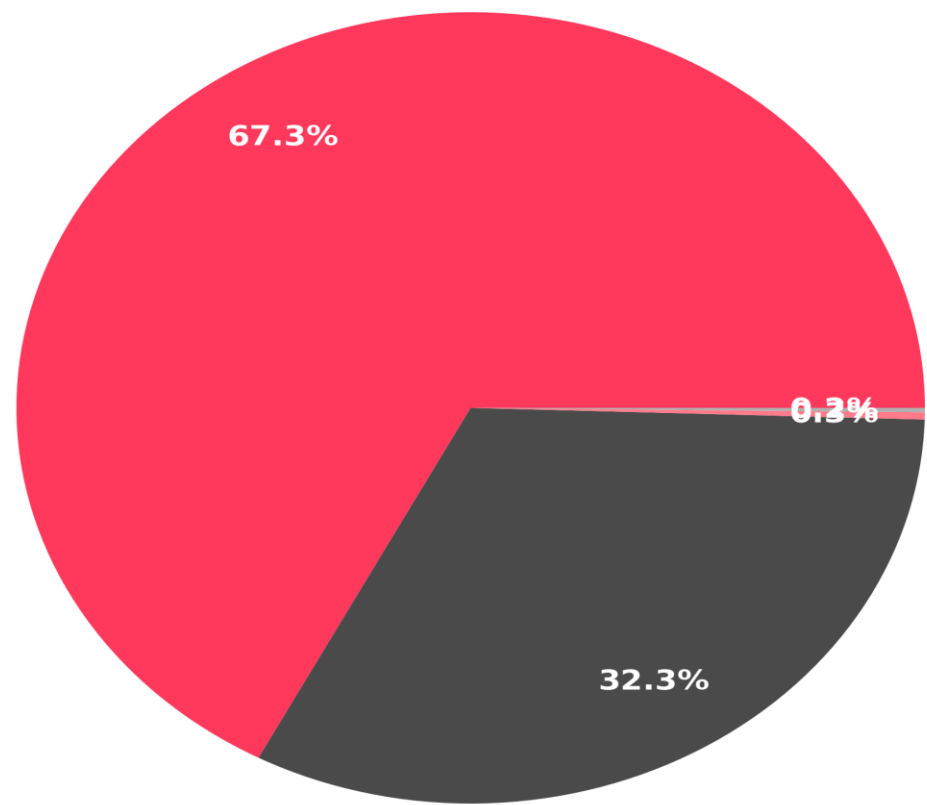
Most listed borough

Insight

Westminster leads supply with 8,000+ listings — but it isn't the priciest borough, showing that high listing volume doesn't always mean the highest rates.

Room Type Distribution

Listing Distribution by Room Type



- Room Type
- Entire home/apt
 - Private room
 - Shared room
 - Hotel room

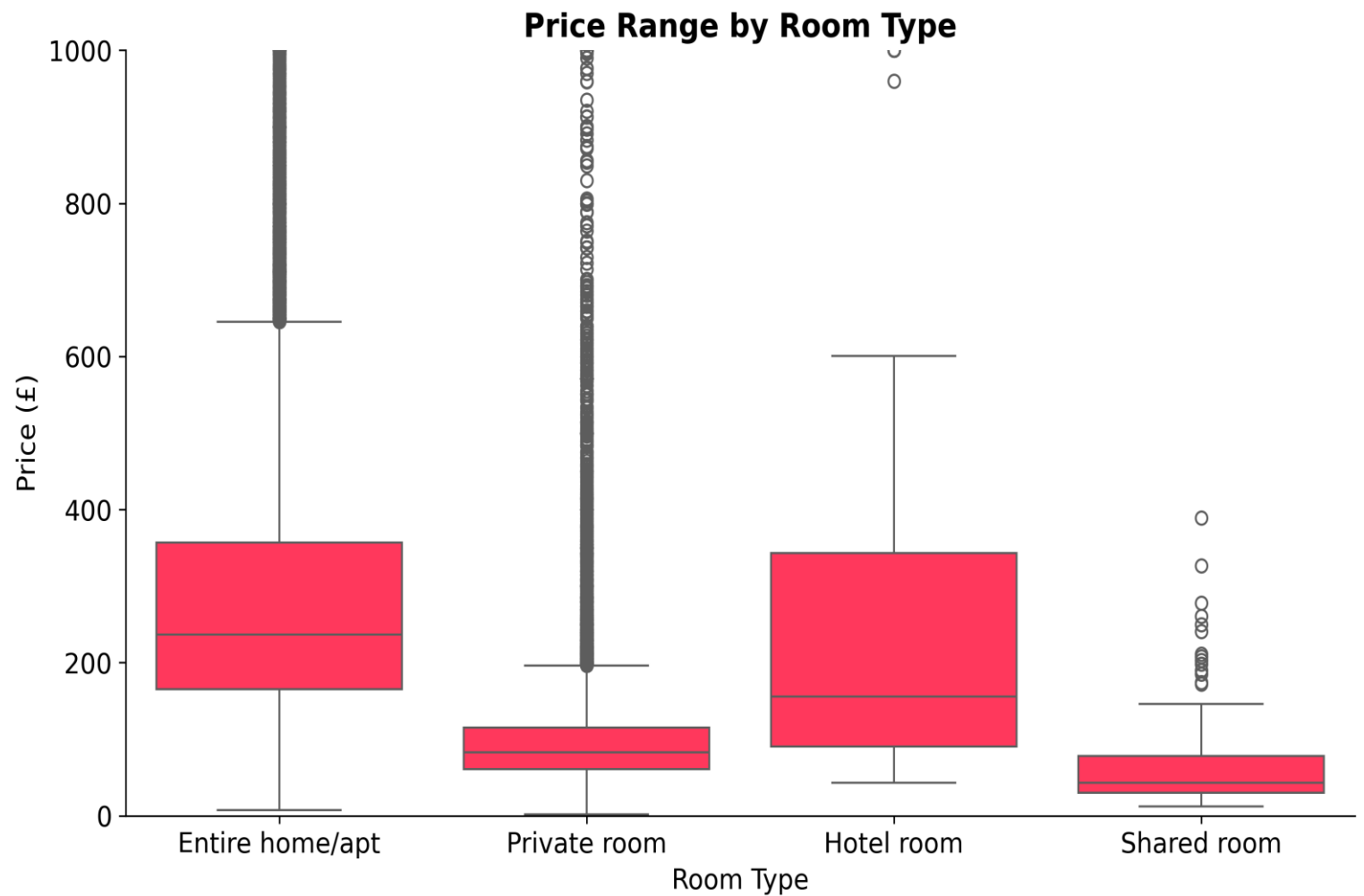
67.3%

Entire home/apt share

Insight

Entire homes/apartments dominate at 67.3% of all listings, while shared and hotel rooms together make up under 1% — London's Airbnb market is overwhelmingly whole-property rentals.

Price by Room Type



Insight

Entire homes carry both a higher median price and a wider price spread than private rooms, confirming room type is one of the strongest price drivers in this market.

Price Distribution



£179

Median nightly price

Insight

Prices are heavily right-skewed: most listings cluster between £80–£300/night, with a long tail of premium listings pulling the average above the median.

Availability vs Price



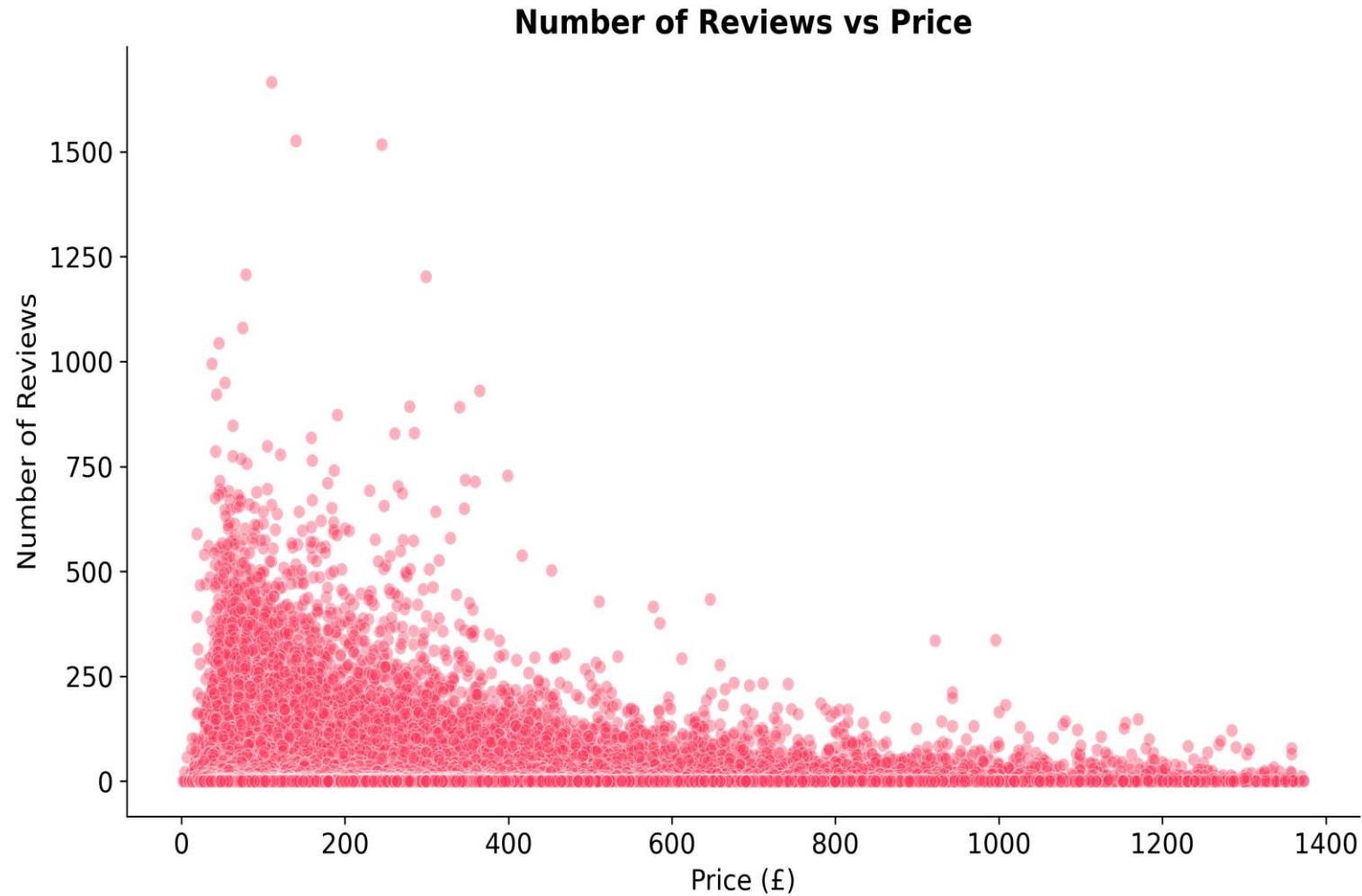
215 days/yr

Avg. availability

Insight

There is only a very weak positive relationship ($r \approx 0.07$) between price and yearly availability — pricing alone does not explain how often a listing is open for booking.

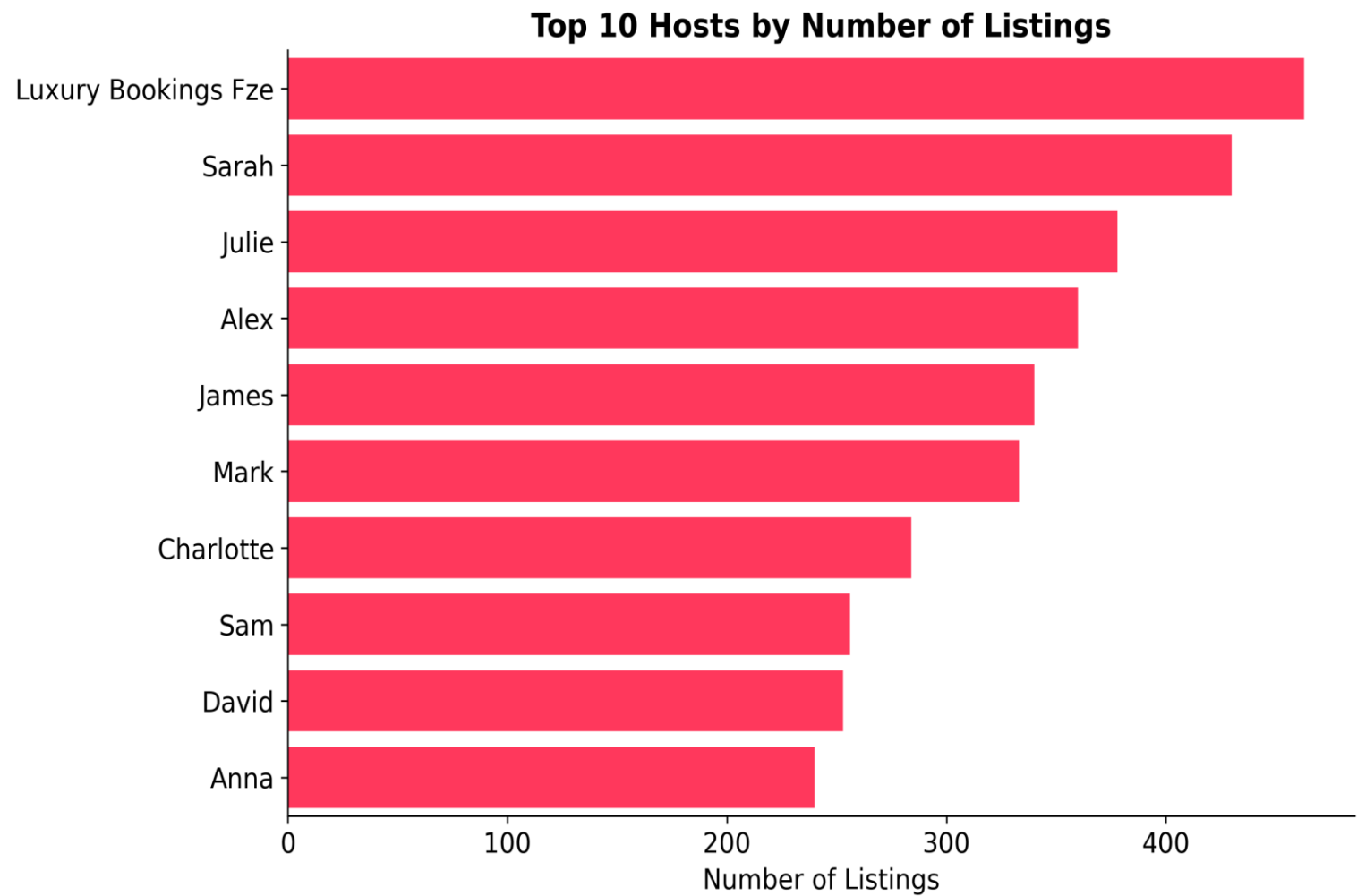
Reviews vs Price



Insight

A weak negative correlation ($r \approx -0.09$) suggests cheaper listings tend to accumulate slightly more reviews — likely reflecting higher booking frequency at lower price points.

Top Hosts by Listings



463

Top host's listings

Insight

The top host manages 463 listings — a clear sign that part of London's Airbnb supply comes from professional, large-scale property management rather than individual hosts.

Key Insights

1

Location drives price more than volume — City of London is priciest, Westminster has the most supply, and they aren't the same borough.

2

Room type is a major price lever — entire homes command higher and more variable prices than private rooms.

3

The market is whole-property dominated — 67.3% of listings are entire homes; shared/hotel rooms are negligible.

4

Price weakly relates to booking activity — availability and reviews both show only weak correlation with price.

5

Professional hosting is present at scale — a top host manages 463 listings, pointing to commercial property management, not just individual homeowners.

Conclusion & Recommendations

For Hosts

Position entire-home listings in high-demand boroughs like City of London or Westminster to maximize pricing power.

For Investors

Evaluate boroughs on both price AND supply — high-price, low-supply boroughs may signal underserved demand.

For Platforms

Weak price-to-booking correlation suggests other factors (photos, reviews, location) drive bookings more than price alone — worth deeper study.

Thank you